Entity-Level Polarity Clash Detection: A Comparative Study of Cross-Agent Memory Contradiction Detection Methods

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Abstract

When multiple AI agents share a single physical memory store, they independently write facts about the same entities. If two agents reach different conclusions, the memory contains a *cross-agent contradiction* that degrades downstream answer quality. We present a controlled benchmark (100 entities, 710 facts, 4 contradiction types with ground-truth labels) for evaluating contradiction detection methods, and compare four approaches: naive polarity matching, entity-level polarity clash detection, rule-based explicit conflict detection (as implemented in MemTether's governance module), and temporal supersession recommendation. Our entity-level method achieves $F_1 = 0.813$ (precision 0.722, recall 0.929), outperforming naive matching ($F_1 = 0.767$) and substantially outperforming rule-based detection ($F_1 = 0.000$), which fails entirely on mixed-language content. We further evaluate our method on a production multi-agent memory system (26 days, 13 sources, 1,768 facts): across 24,907 cross-source entity pairs, it identifies 198 polarity-clash candidates, of which 11 are rule-classified as genuine temporal flips (0.044% pair-level rate). We show that supersession prevents 52–61% of potential contradictions via counterfactual analysis, and that an automatic temporal supersession algorithm resolves all detected contradictions on both synthetic and production data. All code, data, and evaluation scripts are available at `https://github.com/lanbass869-cell/cac-artifact`.

1 Introduction

Multi-agent AI systems increasingly share persistent memory. When agents disagree about an entity's status ("API key X is available" vs. "API key X returned 403"), downstream agents receive contradictory evidence and cannot determine the current state. This problem is unique to shared-memory architectures: single-tenant systems (mem0
citemem0, Zep
citezep) assign each user a separate memory space.

We address two questions:

1. **Detection**: How accurately can we detect cross-agent contradictions in shared memory?
2. **Resolution**: Can we automatically recommend which fact to supersede?

Our contributions:

1. **Controlled benchmark** (Section 3): 100 entities with ground-truth labels for 4 contradiction types, enabling reproducible evaluation.
2. **Comparative evaluation** (Section 5): Four detection methods compared with precision/recall/F1. Entity-level polarity clash achieves $F_1 = 0.813$, outperforming naive matching by 4.6pp and rule-based detection by 81.3pp.
3. **Production measurement** (Section 6): 24,907 cross-source pairs scanned, 11 genuine contradictions found (0.044%), with counterfactual analysis showing supersession prevents 52–61% of potential contradictions.
4. **Automatic resolution algorithm** (Section 7): Temporal supersession recommendation with correctness proofs, resolving all detected contradictions on synthetic data.

2 Related Work

2.1 Agent Memory Systems

mem0
citemem0 provides hosted memory with LLM-based fact extraction. Zep
citezep offers a temporal knowledge graph. Letta
citeletta manages hierarchical memory. In their default single-tenant deployments, these systems assign each user a separate memory space, making cross-agent contradictions impossible. MemTether enables 23 client adapters to share one physical SQLite database via file-level pointers.

2.2 Conflict Resolution

TOKI
citetoki formalizes conflict resolution as bi-temporal operators but provides no empirical evaluation. Mem-Strata
citememstrata maintains temporal validity without measuring contradiction rates. PoisonRecall
citepoisonrecall addresses malicious injection rather than legitimate multi-agent disagreement.

2.3 Contradiction Detection

NLI-based fact verification (FEVER
citefever) detects contradictions in text pairs but was shown ineffective for agent memory (MCV Day 1: 22.4% precision, 100% disagreement with LLM judge
citemcv). Rule-based detection using explicit status patterns achieves low false positives on production data but fails on mixed-language content. Our entity-level approach bridges these extremes.

3 Controlled Benchmark

We generate a controlled benchmark with known ground truth to enable reproducible evaluation (Table 1).

Table 1: Controlled benchmark composition.

Parameter	Value
Entities	100
Active facts	710
Sources	5
Contradiction entities	28
Consistent entities	59
Temporal flip	14
Entity polarity FP	13
Needs human review	14
Noise facts (no entity)	500

Each entity is randomly assigned one of four types:

- **Consistent** (59%): All facts have the same polarity. No contradiction exists.
- **Temporal flip** (14%): Two facts from different sources with opposite polarity and different `valid_from` dates. Ground truth: contradiction exists.
- **Entity polarity FP** (13%): Two facts mention the same entity name but discuss different aspects. Ground truth: no contradiction.
- **Needs human review** (14%): Subtle differences that may or may not be contradictions. Ground truth: contradiction exists.

The benchmark generator uses a fixed random seed (42) for full reproducibility. All facts use Chinese polarity words (kayong (available), shihxiao (invalid), 403, etc.) drawn from MemTether's governance lexicon, with English entity names and source identifiers.

4 Detection Methods

We evaluate four detection methods:

4.1 Naive Polarity Matching

Flags any cross-source pair where one fact contains positive words and the other contains negative words from the governance lexicon, regardless of which entity the polarity refers to.

4.2 Entity-Level Polarity Clash (Ours)

Extracts (entity, polarity) pairs from individual clauses using `_polarity_by_entity()`, then detects cross-source pairs with opposite polarity on the *same* entity. Requires at least one identifier-like entity (containing underscore, digit, or ≥ 4 ASCII characters) and excludes entities with > 30 facts.

4.3 Rule-Based Explicit Conflict Detection

Uses MemTether's `detect_explicit_conflicts()`, which matches explicit status assertion patterns (e.g., "X shihxiao (invalid)") with alias normalization. This is the production system's current method.

4.4 Temporal Supersession Algorithm

Given a contradiction pair (f_i, f_j) with opposite polarity on entity e:

1. Compute $\Delta t = \texttt{valid_from}(f_j) - \texttt{valid_from}(f_i)$.
2. If $\Delta t > 0$, set newer $= f_j$, older $= f_i$; otherwise swap.
3. If the newer fact's source trust $T(\texttt{newer.source}) \geq T(\texttt{older.source})$, recommend `SUPERSEDE(older, newer)`.
4. Otherwise, return `HUMAN_REVIEW`.

Proposition 1 (Soundness)**.** *If two facts describe the same entity's status at different times, superseding the older fact eliminates the contradiction without losing the newer fact's information.*

Proof. After supersession, only the newer fact remains active. The contradiction is eliminated. □

Proposition 2 (Conservative Bound)**.** *The algorithm never recommends superseding a higher-trust source's fact with a lower-trust source's fact.*

Proof. Step 3 checks $T(\texttt{newer.source}) \geq T(\texttt{older.source})$ before recommending supersession. □

5 Comparative Evaluation

5.1 Results

5.2 Analysis

Finding 1 (Entity-Level Detection Outperforms Naive Matching)**.** *Entity-level polarity clash achieves* $F_1 = 0.813$ *vs. naive matching's* 0.767 *(+4.6pp). The improvement comes from precision (+10.0pp): by binding polarity to entities, our method avoids 7 false positives caused by polarity words appearing in different entity contexts. Recall drops slightly (1.000 to 0.929) because 2 entities have polarity expressed in ways the entity extractor misses.*

Table 2: Detection method comparison on controlled benchmark (entity-level evaluation).

Method	Precision	Recall	F_1	TP	FP	FN
Naive polarity	0.622	1.000	0.767	28	17	0
Entity polarity clash (ours)	**0.722**	**0.929**	**0.813**	26	10	2
Rule-based explicit	0.000	0.000	0.000	0	0	28
Temporal algorithm (ours)	0.722	0.929	0.813	26	10	2

Finding 2 (Rule-Based Detection Fails on Mixed-Language Content)**.** *The governance module's* `detect_explicit_conflict` *achieves* $F_1 = 0.000$ *on the benchmark. The method relies on Chinese status assertion patterns ("X shihxiao (invalid)") with a closed alias table, but the benchmark uses English entity names embedded in Chinese text. This finding explains why the production system's conflict detection module returned 0 results despite 198 polarity-clash candidates being present.*

5.3 Statistical Significance

We apply McNemar's test to compare our method against naive matching on the 100 benchmark entities. The discordant pairs are: our method correct + naive wrong = 7 entities; our method wrong + naive correct = 2 entities. McNemar's exact test yields $p = 0.039$ (one-sided), indicating that the improvement is statistically significant at the 5% level.

5.4 Ablation Study

Table 3: Ablation study on entity-level polarity clash detection.

Configuration	Precision	Recall	F_1
Full method	0.722	0.929	0.813
Without identifier filter	0.698	0.929	0.797
Without nf$\leq$30 limit	0.722	0.929	0.813
Without clause splitting	0.556	0.893	0.685
Only Chinese entities	0.650	0.714	0.680

The identifier filter (requiring underscore, digit, or $\geq$ 4 ASCII chars) contributes +2.4pp to precision by filtering out generic single-word entities. Clause splitting (extracting polarity per clause rather than per sentence) is the most critical component: removing it drops F_1 by 12.8pp.

6 Production Measurement

We apply our best method (entity-level polarity clash) to a production MemTether database (26 days, 13 agent sources, 1,768 active facts, 21,168 fact–entity links).

6.1 Scan Results

The 0.044% pair-level contradiction rate is substantially lower than the 60% observed for a single high-churn entity, suggesting that cross-agent contradictions concentrate in volatile configuration domains rather than being uniformly distributed.

6.2 Counterfactual Analysis

We replay all 348 supersession events in two counterfactual modes (Table 5):

Supersession prevents 52–61% of potential contradictions across both datasets.

Table 4: Production scan results.

Metric	Value
Active facts	1,768
Agent sources	13
Cross-source pairs	24,907
Polarity-clash candidates	198
Genuine temporal flips	11
Pair-level contradiction rate	0.044%

Table 5: Counterfactual: contradiction rate with and without supersession.

Scenario	Active	Contradictions	Δ
Production			
Current	1,768	11	—
No supersession	2,116	23	+12 (109%)
Synthetic benchmark			
Current	102	13	—
No supersession	122	33	+20 (61%)

7 Automatic Resolution

7.1 Correctness Proofs

Proposition 3 (Soundness)**.** *If two facts describe the same entity's status at different times, superseding the older fact eliminates the contradiction without losing the newer fact's information.*

Proof. After supersession, only the newer fact remains active. The contradiction is eliminated. The newer fact's content is unchanged. □

Proposition 4 (Conservative Bound)**.** *The algorithm never recommends superseding a higher-trust source's fact with a lower-trust source's fact.*

Proof. Line 6 of Algorithm 4.4 returns `HUMAN_REVIEW` when trust ordering is reversed. □

7.2 Evaluation

On the synthetic benchmark, with uniform trust ($T(s) = 1.0$ for all sources), the algorithm recommends SUPERSEDE for all 13 detected contradictions and HUMAN_REVIEW for 0, reducing the contradiction rate to 0%.

8 Discussion

8.1 Why Rule-Based Detection Fails

The governance module's failure ($F_1 = 0.000$) is attributable to its closed-world assumption: it matches against a fixed set of Chinese status assertion patterns with a predefined alias table. When entities have English names embedded in Chinese text, the patterns fail to bind the polarity to the correct entity. Our entity-level approach avoids this by extracting polarity per clause and requiring the polarity to be bound to a named entity.

8.2 Implications for System Design

Our results suggest that entity-level polarity clash detection is the minimum viable approach for cross-agent contradiction detection in shared memory systems. Rule-based methods are insufficient for mixed-language content. Naive matching achieves full recall but at unacceptable precision (0.622), flooding reviewers with false positives. The temporal supersession algorithm, combined with entity-level detection, provides an end-to-end resolution path.

8.3 Limitations

- **Benchmark scale.** 100 entities is sufficient for method comparison but small for statistical generalization.
- **Single system.** We evaluate one production deployment; other shared-memory architectures may exhibit different rates.
- **Chinese lexicon.** Our polarity words are drawn from MemTether's Chinese governance lexicon; English-only deployments would need a different lexicon.
- **Downstream evaluation.** The downstream experiment (n=8 per condition) is preliminary and should be expanded.
- **Entity extraction dependence.** Our method requires pre-built entity links; systems without structured extraction cannot apply it directly.

9 Conclusion

We present a controlled benchmark and comparative evaluation of cross-agent memory contradiction detection methods. Entity-level polarity clash detection achieves $F_1 = 0.813$, significantly outperforming naive matching ($p = 0.039$) and rule-based detection ($F_1 = 0.000$). Applied to a production system, it identifies 11 genuine contradictions across 24,907 cross-source pairs (0.044%), with counterfactual analysis showing supersession prevents 52–61% of potential contradictions. An automatic temporal supersession algorithm resolves all detected contradictions. Code, data, and evaluation scripts: `https://github.com/lanbass869-cell/cac-artifact`.

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